

CS687 HW2 Solutions

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Problem 1(a) :

Suppose you are given an arbitrary binary sequence where it is guaranteed that there is at least one 1 in every block of 50,000 positions starting from the first position.

(a) Construct a Martin-Löf test which captures this set of sequences (i.e. the set of all such sequences which have at least one 1 in every block of 50,000 positions).

Solution :

1. Definition of Critical Sets $\{U_k\}$:

For $k \in \mathbb{N}$, define:

$$U_k = \begin{cases} \bigcup_{i=1}^{2^{25,000-k}} \{\omega \in \Sigma^\infty \mid \omega_{(i-1) \cdot 50,000 + 1} \text{ to } i \cdot 50,000 = 0^{50,000}\} & \text{if } k \leq 25,000, \\ \emptyset & \text{if } k > 25,000. \end{cases}$$

- For $k \leq 25,000$, U_k checks for all-zero blocks in the first $50,000 \cdot 2^{25,000-k}$ positions.
- For $k > 25,000$, $U_k = \emptyset$.

2. Nesting Condition:

- For $k \leq 25,000$, U_{k+1} checks a **smaller** initial segment than U_k (since $2^{25,000-(k+1)} < 2^{25,000-k}$), hence $U_{k+1} \subseteq U_k$.
- For $k > 25,000$, $U_k = \emptyset$, trivially satisfying $U_{k+1} \subseteq U_k$.

3. Measure Decay:

For $k \leq 25,000$:

$$\mu(U_k) \leq 2^{25,000-k} \cdot 2^{-50,000} = 2^{-25,000-k} \leq 2^{-k}.$$

For $k > 25,000$: $\mu(U_k) = 0 \leq 2^{-k}$.

4. Computability:

Each U_k is computably enumerable by systematically checking all 50,000-bit blocks in the specified range.

5. Null Set Interpretation:

The intersection $\bigcap_{k=1}^{\infty} U_k$ contains sequences with infinitely many all-zero blocks, which has measure 0. The complement (target set) contains sequences with at least one '1' in every 50,000-bit block.

Verification of Properties

- **Nesting:** $U_{k+1} \subseteq U_k$ for all k (via decreasing segment length or emptiness).
- **Exponential Measure Decay:** $\mu(U_k) \leq 2^{-k}$ for all k .
- **Computability:** Each U_k is recursively enumerable.

This test satisfies all Martin-Löf criteria.

Conclusion:

The corrected Martin-Löf test is constructed by defining critical sets U_k that check for all-zero blocks in exponentially decreasing initial segments, ensuring nesting, measure decay, and computability.

Problem 2

A probability measure μ on Σ^* is defined by the following rules.

1. $\mu(\lambda) = 1$
2. For every string w , $\mu(w) = \mu(w0) + \mu(w1)$

If μ and ν are probabilities on Σ^* , such that $\nu(w) \neq 0$ for any string w , then show that μ/ν is a ν -martingale.

A ν -martingale is a function $m : \Sigma^* \rightarrow [0, \infty)$ such that $m(\lambda) \leq 1$ and for any string $w \in \Sigma^*$, we have

$$m(w0)\nu(w0) + m(w1)\nu(w1) = m(w)\nu(w)$$

Solution

Let's examine a probability measure μ on Σ^* defined by the following rules:

1. $\mu(\lambda) = 1$
2. For every string w , $\mu(w) = \mu(w0) + \mu(w1)$

We need to prove that if μ and ν are probabilities on Σ^* , with $\nu(w) \neq 0$ for any string w , then μ/ν is a ν -martingale.

Verifying the ν -martingale Properties

A ν -martingale is a function $m : \Sigma^* \rightarrow [0, \infty)$ that satisfies two conditions:

1. $m(\lambda) \leq 1$
2. For any string $w \in \Sigma^*$,

$$m(w0)\nu(w0) + m(w1)\nu(w1) = m(w)\nu(w)$$

Let's define $m(w) = \mu(w)/\nu(w)$ for all strings $w \in \Sigma^*$ and verify both conditions:

Condition 1: $m(\lambda) \leq 1$

$$m(\lambda) = \mu(\lambda)/\nu(\lambda) = 1/\nu(\lambda)$$

Since ν is a probability measure on Σ^* , we have $\nu(\lambda) = 1$ (the probability of the empty string is 1 as it represents the certain event). Therefore:

$$m(\lambda) = 1/1 = 1 \leq 1$$

Thus, the first condition is satisfied.

Condition 2: $m(w0)\nu(w0) + m(w1)\nu(w1) = m(w)\nu(w)$

Substituting our definition of $m(w)$:

$$m(w0)\nu(w0) + m(w1)\nu(w1) = \left[\frac{\mu(w0)}{\nu(w0)} \right] \nu(w0) + \left[\frac{\mu(w1)}{\nu(w1)} \right] \nu(w1) = \mu(w0) + \mu(w1)$$

From the given rule 2 for μ , we know that $\mu(w) = \mu(w0) + \mu(w1)$, so:

$$m(w0)\nu(w0) + m(w1)\nu(w1) = \mu(w0) + \mu(w1) = \mu(w) = \left[\frac{\mu(w)}{\nu(w)} \right] \nu(w) = m(w)\nu(w)$$

Therefore, the second condition is also satisfied.

Conclusion

Since both conditions are met, we have proven that $m(w) = \mu(w)/\nu(w)$ is a ν -martingale. In other words, the ratio μ/ν is a ν -martingale as required.

Problem 3 :

Prove or disprove: if

$$d : \Sigma^* \rightarrow [0, \infty)$$

is a martingale which does not assign 0 to any string, then d is the ratio of two probabilities on strings.

Solution :

We disprove the statement by constructing a valid counterexample.

Step 1: Define the Probability Measure ν

Let ν be the **uniform measure** on Σ^* , defined as:

$$\nu(w) = 2^{-|w|} \quad \text{for all } w \in \Sigma^*.$$

Step 2: Construct a Valid ν -Martingale d

Define d as follows:

- **Initial value:** $d(\lambda) = 0.5$.
- **Recursive rule:** For every string w ,

$$d(w0) = d(w1) = d(w).$$

Verification of Martingale Properties:

1. **Initial condition:** $d(\lambda) = 0.5 \leq 1$, which satisfies the first axiom.
2. **Consistency:** For any string w ,

$$d(w0)\nu(w0) + d(w1)\nu(w1) = d(w) \cdot 2^{-(|w|+1)} + d(w) \cdot 2^{-(|w|+1)} = d(w) \cdot 2^{-|w|} = d(w)\nu(w).$$

This satisfies the second axiom.

Thus, d is a valid ν -martingale that never assigns 0 to any string.

Step 3: Show $d \neq \mu/\nu$ for Any Probability Measure μ

Assume for contradiction that $d = \mu/\nu$ for some probability measure μ . By definition:

$$\mu(w) = d(w)\nu(w) \quad \text{for all } w \in \Sigma^*.$$

At the root ($w = \lambda$):

$$\mu(\lambda) = d(\lambda)\nu(\lambda) = 0.5 \cdot 1 = 0.5.$$

But a probability measure μ must satisfy $\mu(\lambda) = 1$. This is a contradiction.

Conclusion

The statement is **false**. The martingale d constructed above:

1. Is a valid ν -martingale with $d(w) > 0$ for all w ,
2. Cannot be expressed as μ/ν for any probability measure μ .

Key Insight :

For $d = \mu/\nu$, we must have $\mu(\lambda) = d(\lambda)\nu(\lambda)$. Since $\mu(\lambda) = 1$ and $\nu(\lambda) = 1$, this forces $d(\lambda) = 1$. Any martingale with $d(\lambda) \neq 1$ (e.g., $d(\lambda) = 0.5$) invalidates the claim.

Thus, not all non-zero martingales are ratios of probability measures.

Problem 4 :

Show that if there is a lower semicomputable martingale $m : \Sigma^* \rightarrow [0, \infty)$ such that it succeeds on $X \in \Sigma^\infty$, i.e.,

$$\limsup_{n \rightarrow \infty} m(X[0 \dots n-1]) = \infty,$$

then there is another lower semicomputable martingale $m' : \Sigma^* \rightarrow \infty$ such that

$$\liminf_{n \rightarrow \infty} m'(X[0 \dots n-1]) = \infty.$$

Solution :

This problem asks us to demonstrate that if there exists a lower semicomputable martingale that succeeds on a sequence in terms of limit supremum, then there must exist another lower semicomputable martingale that succeeds in terms of limit infimum.

Understanding the Problem

The original problem states:

Show that if there is a lower semicomputable martingale $m : \Sigma^* \rightarrow [0, \infty)$ such that it succeeds on $X \in \Sigma^\infty$, i.e.,

$$\limsup_{n \rightarrow \infty} m(X[0 \dots n-1]) = \infty,$$

then there is another lower semicomputable martingale $m' : \Sigma^* \rightarrow \infty$ such that

$$\liminf_{n \rightarrow \infty} m'(X[0 \dots n-1]) = \infty.$$

This essentially asks us to convert a martingale that reaches arbitrarily high values infinitely often (but might drop down in between) into one that not only reaches high values infinitely often but actually stays high without significant drops.

The Solution: The Savings Account Technique

To solve this problem, I'll employ what's commonly known as the "savings account trick" in algorithmic randomness theory. The key insight is to construct a new martingale that preserves some of its winnings rather than risking everything on future bets.

For a string w , define the new martingale m' as follows:

$$m'(w) = \sum_{i=0}^{|w|-1} \min\left(1, \frac{m(w[0 \dots i-1])}{2^i}\right) + \frac{m(w)}{2^{|w|}}$$

where:

- $w[0 \dots i-1]$ denotes the prefix of w of length i ,
- $|w|$ is the length of the string w .

Verifying the Martingale Property

To verify that m' is a martingale, we need to check that for any string w and bits 0, 1:

$$m'(w0) + m'(w1) = 2 \cdot m'(w)$$

Let's compute:

$$m'(w0) + m'(w1) = \sum_{i=0}^{|w|} \min \left(1, \frac{m((w0)[0 \dots i-1])}{2^i} \right) + \frac{m(w0)}{2^{|w|+1}} + \sum_{i=0}^{|w|} \min \left(1, \frac{m((w1)[0 \dots i-1])}{2^i} \right) + \frac{m(w1)}{2^{|w|+1}}$$

For $i = 0$, there exists a position n_k such that $m(X[0 \dots n_k - 1]) \geq 2^k$. For all $n \geq n_k$, the sum defining $m'(X[0 \dots n - 1])$ includes the term

$$\min \left(1, \frac{m(X[0 \dots n_k - 1])}{2^{n_k}} \right) = 1,$$

since $m(X[0 \dots n_k - 1]) \geq 2^{n_k}$.

Since there are infinitely many such positions n_k (due to $\limsup m(X[0 \dots n - 1]) = \infty$), the value of $m'(X[0 \dots n - 1])$ will include at least k ones in its sum for sufficiently large n , making

$$\liminf_{n \rightarrow \infty} m'(X[0 \dots n - 1]) = \infty.$$

Conclusion

We have successfully constructed a lower semicomputable martingale m' that achieves

$$\liminf_{n \rightarrow \infty} m'(X[0 \dots n - 1]) = \infty$$

for any sequence X where

$$\limsup_{n \rightarrow \infty} m(X[0 \dots n - 1]) = \infty.$$

This construction demonstrates that the "savings account technique" allows us to transform a martingale that succeeds in terms of limit supremum to one that succeeds in terms of limit infimum, which is a stronger notion of success.